Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — graduate level (Listas 1–2)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Price indices, PPP and welfare across countries

(Kurlat chs. 1–2 • 25 points)

Question 1. (10 points) Objective items. Each item needs a short derivation; write it on scratch paper.

a) (3 points) A household has preferences $u(x_1, x_2) = x_1^{1/3} x_2^{2/3}$ over electricity (x_1) and everything else (x_2). Between two months the price of electricity rises 44% and the other price does not change. The true cost-of-living (Konüs) index, defined as the ratio of minimum expenditures needed to reach the *same* utility at the new and old prices, records inflation of:

A. 14.67%

B. 12.92%

C. 12.99%

D. 11.34%

b) (3 points) Brazil and the US produce a tradable good with $Y_T = A_T L_T$ and a non-tradable service with $Y_N = A_N L_N$. Labor is mobile across sectors within each country, markets are competitive, the tradable obeys the law of one price, and consumers' price index is $P = p_T^{1/2} p_N^{1/2}$. Brazil's productivity relative to the US is 0.25 in tradables and 0.80 in non-tradables. Brazil's price level relative to the US, both measured in dollars at the market exchange rate, is:

A. 0.250

B. 0.800

C. 0.559

D. 0.313

c) (2 points, T or F) With Cobb–Douglas preferences the Fisher index equals the true cost-of-living index, since it is the geometric mean of an upper bound (Laspeyres) and a lower bound (Paasche).

d) (2 points, T or F) If log consumption is normal with variance v and utility is CRRA with curvature σ, the consumption-equivalent cost of inequality, $1 - e^{-\sigma v/2}$, depends neither on mean consumption nor on the value of life $\bar{u}$, while the welfare cost of shorter life expectancy depends on both.

Question 2. (15 points) Discursive. The Jones–Klenow measure (Kurlat §2.2). "Rawls" is placed at random in country $j \in \{BR, US\}$. He is alive with probability a_j (life expectancy over 100) and, if alive, gets $\bar{u} + u(c) - \theta(1 - l_j)^2$, where $1 - l_j$ is market work (the same for everyone) and $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ ($\ln c$ if $\sigma = 1$). Consumption is lognormal: $\ln c \sim N(m_j, v_j)$, with mean $\bar{c}_j = e^{m_j + v_j/2}$. The welfare of Brazil relative to the US is the λ that makes Rawls indifferent between Brazil and a US in which *every* consumption level is multiplied by λ. Data: $\bar{c}_{US} = 1$, $\bar{c}_{BR} = 0.30$, $v_{US} = 0.4$, $v_{BR} = 1.0$, $a_{US} = 0.79$, $a_{BR} = 0.76$, $1 - l_{US} = 0.20$, $1 - l_{BR} = 0.22$, $\theta = 2$, $\bar{u} = 5$.

a) (3 points) Prove that $\mathbb{E}[u(c)] = u(c_{CE})$ with $c_{CE} = \bar{c}\, e^{-\sigma v/2}$ for every $\sigma > 0$, including $\sigma = 1$. Interpret the factor $e^{-\sigma v/2}$.

Solution:

b) (4 points) Let $\sigma = 1$. Derive λ and an *exact* additive decomposition of $\ln \lambda$ into consumption, inequality, leisure and life-expectancy terms. Compute every term.

Solution:

c) (4 points) Now let σ be general. Derive λ in closed form. Prove that it tends to the formula of (b) as $\sigma \to 1$. Prove that the pure inequality factor (all else equal across countries) falls with σ when $v_{BR} > v_{US}$, and give its limits as $\sigma \to 0$ and $\sigma \to \infty$. Explain why Kurlat's unnormalized $c^{1-\sigma}/(1-\sigma)$ with a fixed $\bar{u}$ has no limit at $\sigma = 1$.

Solution:

d) (4 points) Compute the inequality factor at $\sigma = 1$ and $\sigma = 2$. Then show that with $\sigma = 2$ and $\bar{u} = 5$ the flow utility of the Brazilian certainty-equivalent resident is negative, and explain why that makes λ meaningless. What must the analyst do?

Solution:

Block 2 — Solow: CES, exact dynamics and human capital

(Kurlat chs. 4–5; Romer ch. 1 • 25 points)

Question 3. (10 points) Objective items. Work *per effective worker*: $\tilde{k} = K/(AL)$, $\dot{\tilde{k}} = sf(\tilde{k}) - (n+g+\delta)\tilde{k}$ in continuous time.

a) (3 points) Let $f(\tilde{k}) = [\alpha \tilde{k}^{\rho} + 1 - \alpha]^{1/\rho}$ with $\alpha = 1/3$, $\rho = 1/2$ (elasticity of substitution 2), and $n + g + \delta = 0.08$. The saving rate above which $\tilde{k}$ grows forever, with no steady state, is:

A. 24%

B. 33.3%

C. 72%

D. none: the Solow model always has a steady state

b) (3 points) Let $f(\tilde{k}) = \tilde{k}^{1/3}$ and $n + g + \delta = 0.08$. The economy starts at $\tilde{k}(0) = \tilde{k}^*/8$. The time until $\tilde{k}$ reaches $\tilde{k}^*/2$ is:

A. 13.0 years

B. 20.6 years

C. 13.2 years

D. 6.6 years

c) (2 points, T or F) With CES production and an elasticity of substitution below one ($\rho < 0$), the capital share $\alpha_K(\tilde{k}) = f'(\tilde{k})\tilde{k}/f(\tilde{k})$ falls as $\tilde{k}$ rises.

d) (2 points, T or F) On a balanced growth path with $Y = K^{\alpha}(AL)^{1-\alpha}$ and A growing at g, growth accounting attributes all of the growth of output per worker to the Solow residual.

Question 4. (15 points) Discursive. Mankiw–Romer–Weil with Brazilian data. Output is

$$Y = K^{\alpha} H^{\beta} (AL)^{1-\alpha-\beta}, \qquad \alpha + \beta < 1,$$

with $\dot{K} = s_k Y - \delta K$ and $\dot{H} = s_h Y - \delta H$; L and A grow at n and g. Let $k = K/(AL)$, $h = H/(AL)$, $y = k^{\alpha} h^{\beta}$ and $N \equiv n + g + \delta$.

a) (3 points) Derive the laws of motion of k and h, the steady state (k^*, h^*) and $\ln y^*$. Give the elasticity of y^* with respect to s_k.

Solution:

b) (4 points) Prove local stability. Linearize the system in $x = \ln k$, $z = \ln h$, compute the Jacobian at the steady state, its eigenvalues, and the speed at which $\ln y$ converges. Compare with the textbook Solow model ($\beta = 0$) for $\alpha = \beta = 1/3$, $N = 0.06$.

Solution:

c) (4 points) Find the saving rates (s_k, s_h) that maximize steady-state consumption per effective worker $c^* = (1 - s_k - s_h)y^*$, and prove that they are a global maximum. What happens to the convergence speed and to y^* as $\alpha + \beta \to 1$, and what is the case $\beta \to 0$?

Solution:

d) (4 points) Use $\alpha = \beta = 1/3$; Brazil: $s_k = 0.18$, $s_h = 0.10$, $N = 0.065$; US: $s_k = 0.22$, $s_h = 0.12$, $N = 0.06$. Compute the predicted ratio y^*_{BR}/y^*_{US}. Output per worker is $A \cdot y$; if the observed ratio of output per worker is 0.25, compute the ratio of A and of Hicks-neutral TFP $B \equiv A^{1-\alpha-\beta}$. Split the log gap into accumulation and technology, and say what the model predicts for the gap's dynamics.

Solution: